

Malvern Madondo

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EDUCATION

Emory University, Atlanta, GA

Aug/19 – present

PhD Student, Computer Science & Informatics

Research Interests: Deep Learning; Numerical Optimization; Differential Equations; Chaos Theory; Education Technology;

Coursework: Numerical Methods for Deep Learning*, Database Systems*, Machine Learning*, Intro to Nonlinear

Optimization*, Numerical Analysis I, Algorithms, Artificial Intelligence.

* Current course

The College of Saint Scholastica (CSS), Duluth, MN

Sept/15 – May/19

B.S. in Computer Science and B.A. in Mathematics (*summa cum laude*)

EXPERIENCE

Marketing Department, CSS — Web Assistant, Duluth, MN

Oct/16 – May/19

- Supported the web and marketing goals of the Interactive Services Team through web content design, creation, and editing, while following branding guidelines and best practices. Prepared site pages and content to meet accessibility guidelines.

Facebook — Software Engineering Intern, Menlo Park, CA

May/18 – Aug/18

- Worked with the Crisis Response/Safety Check team. Full-stack and back-end work in Hack, XHP, and JavaScript.
- Worked on bringing Community Help, which allows users to find and offer help during a crisis, to Facebook's mobile site.
- Built a Mark Safe QP on WWW platform that resulted in a 25% increase in primary clicks, 43% increase in QP dismissals and 50% increase in 'Does Not Apply' clicks.

Facebook — Software Engineering Intern, Menlo Park, CA

Jun/17 – Aug/17

- Participated in the Facebook University for Engineering – iOS Development track.
- Developed from scratch an iOS app that integrated location-based facts, Tinder-style fact sharing, localization, image recognition (utilizing IBM Watson API) and Wikipedia descriptions.

State of Minnesota, DHS — Urban Scholar and Management Analyst, St. Paul, MN

May/16 – Aug/16

- Assisted the Member Provider Services communications team in the development and implementation of a process to archive Provider Manual web page revisions and index web page documents.
- Implemented the Minnesota State Government's Plain Language initiative in web page revisions to promote accessibility.
- Participated in the Urban Scholars Leadership Institute. Awarded Certificate of Commendation.

IT Help Desk, CSS — Team Leader and Technology Assistant, Duluth, MN

Sept/15 – Oct/18

- Co-led and supervised a team of 15 TAs offering technical support, troubleshooting and diagnosing network and technical issues.
- Demonstrated advanced customer service skills while interacting and communicating with peers, faculty, and staff from diverse backgrounds.

TECHNICAL SKILLS

Software Development: *Most Experience:* Java, Python, (Keras, Tensorflow, Theano), Hack, Swift, PHP, HTML/CSS;

Some Experience: JavaScript, React, C++, SQL, MATLAB

Tools & Methodologies: Agile Methodology, Project Management, Linux, Git, Mercurial, Maple

Areas: Machine Learning (Scikit-Learn, Keras, Theano, & Tensorflow), Mobile Development (iOS & Android), Web Development.

RESEARCH

Math Capstone Project: Mathematical Foundations of Machine Learning — CSS, Duluth, MN

Sept/18 – Dec/18

- Theoretical research aimed at understanding the core principles of various Machine Learning algorithms.

Student Research Lead for Interactive Services Web Team — CSS, Duluth, MN

Sept/17 – Dec/17

- Web accessibility research to implement accessible accordions and videos in the school's website.

Independent Study on TSP with Prof. Glesener — CSS, Duluth, MN

Jan/17 – May/17

- Learned computational complexity theory and investigated the Traveling Salesman Problem and different algorithms to solve it.
- With 5 other students, developed an optimal path for Campus Security to efficiently check all classrooms and offices on campus.

Research Assistant for Prof. Gibbons — CSS, Duluth, MN

Sept/16 – Dec/16

- Explored keyboard teleoperation and SLAM map building with a ROS turtlebot to have it autonomously navigate a building floor.

PUBLICATIONS

Madondo, M. and Gibbons, T. "Learning and Modeling Chaos Using LSTM Recurrent Neural Networks". Midwest Instruction & Computing Symposium (MICS 2018). http://micsymposium.org/mics2018/proceedings/MICS_2018_paper_26.pdf

Madondo, M., Gomez, DM., and Ciernia, N. "Hitchhiker's Guide to Computer Science for Social Good". Midwest Instruction & Computing Symposium (MICS 2018). http://micsymposium.org/mics2018/proceedings/MICS_2018_paper_41.pdf

TEACHING and MENTORING EXPERIENCE

- CS Department, Emory University** — *Teaching Assistant*, Atlanta, GA *Aug/19 – present*
- Supporting students taking the Intro to CS course by providing 1-on-1 instruction during lectures and holding office hours.
- Math Department, CSS** — *Teaching Assistant*, Duluth, MN *Oct/18 – May/19*
- Supporting students taking Math courses, e.g. Calculus & Linear Algebra, by providing *1-on-1* instruction and holding office hours.
- CodePath.Org** — *Technical Program Manager (iOS Dev Tutor)*, Duluth, MN *Sept/17 – May/19*
- Provide technical support for students during labs and assignments. Lead tutorials and manage in-class activities.
- Google IgniteCS Program** — *Mentor*, Lincoln Park Middle School, Duluth, MN *Jan/18 – May/18*
- Mentored Middle School students and taught weekly course in Computational Thinking as a foundation for Computer Science.

LEADERSHIP and OTHER ACTIVITIES

- NORTHFORCE** — *Student Ambassador (Mentor Connection Program)*, Duluth, MN *Oct/16 – May/19*
- Connect and network with local professionals to learn about career options and gain real world insights.
- Marketing Department (Student Life)** — *Blogger*, Duluth, MN *Sept/15 – May/19*
- Maintain department's social media presence to engage current and prospective students.
 - Provide insights into student life while sharing experiences as a student.
- WellU Program, CSS** — *Storm's Advocate*, Duluth, MN *Aug/16 – May/18*
- Present educational programs and table at events promoting a lifestyle of health and wellness to college community.
- Midwest Instruction & Computing Symposium** — *Robotics Competition* *Apr/16, Apr/17*
- Designed and programmed an autonomous robot that could trace a path and play a 3-hole miniature golf course, with the help of two other students and advice from a faculty.
- Digi-Key Electronics** — *Programming Competition*, Thief River Falls, MN *Oct/17*
- Mentored by Kris Glesener and in a team of four, placed fifth place overall. Won first place in two “blender” sessions.
- Office of International Programs** — *Student Ambassador* *Aug/16 – Sept/16*
- Assisted in the organization and implementation of the summer orientation program for incoming international students.
- CSS Youth Theology Institute** — *Peer Leader* *July/16*
- Mentored participants and led opportunities for community building and small group discussions.
- Mathematical Association of America, NCS** — *Team Competition* *Nov/15, 16, 17, 18*
- Participated in the annual competition and worked on moderately easy to Putnam level Math problems.

AWARDS

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| • Minnesota High Tech Association (MHTA) Foundation Scholar | <i>Nov/18</i> |
| • Affirm Scholar – ACM Richard Tapia Conference | <i>Aug/18</i> |
| • Quip Diversity Scholarship Runner-up | <i>Jul/18</i> |
| • Student Employee of the Year Award | <i>Apr/18</i> |
| • Code2040 Tech Trek Fellow | <i>Mar/17</i> |
| • COR Leadership Award (CSS) | <i>Mar/17</i> |
| • Minnesota Private Colleges - Phillips Scholar | <i>Feb/17</i> |
| • CSS Mentor Connection Certificate | <i>Apr/17</i> |
| • Minneapolis Urban Scholar Leadership Award | <i>Aug/16</i> |
| • Philanthropic Ventures Foundation Grace Scholar | <i>Aug/15 – May/19</i> |

LANGUAGES

- English (*Full Proficiency*)
- Shona (*Full Proficiency*)